The Digestive System

Human Physiology - Chapter 18

Biology Teaching Resources

December 24, 2025

Outline

18.1 Introduction to the Digestive System

18.2 From Mouth to Stomach

18.3 Small Intestine

18.4 Large Intestine

18.5 Liver, Gallbladder, and Pancreas

18.6 Neural and Endocrine Regulation

18.7 Digestion and Absorption

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18.1 Introduction to the Digestive System

- The **digestive system** (消化系统) transfers nutrients, water, and electrolytes from food into the body
- **Digestion** (消化): chemical breakdown of food molecules through hydrolysis
- The system includes the **gastrointestinal (GI) tract** (胃肠道) and **accessory organs** (辅助器官)

18.1 Introduction to the Digestive System

- Hydrolysis reactions break down complex food molecules into simpler, absorbable units
- Carbohydrates → monosaccharides
- Proteins → amino acids
- Lipids → fatty acids and glycerol

18.1 Introduction to the Digestive System

- The GI tract extends from mouth to anus (approximately 9 meters)
- Includes: mouth, pharynx, esophagus, stomach, small intestine, large intestine
- Accessory organs: salivary glands, liver, gallbladder, pancreas

18.1 Introduction to the Digestive System

- The GI tract wall consists of four layers from innermost to outermost:
  - **Mucosa** (粘膜): epithelium, connective tissue, smooth muscle
  - **Submucosa** (粘膜下层): connective tissue with blood vessels and nerves
  - **Muscularis externa** (肌层): inner circular and outer longitudinal smooth muscle
  - **Serosa** (浆膜): connective tissue with epithelium

18.1 Introduction to the Digestive System

- Regulation of the GI tract involves:
  - **Extrinsic control** (外源性调节): autonomic nervous system
  - **Intrinsic control** (内源性调节): enteric nervous system
  - **Paracrine regulation** (旁分泌调节): local chemical signals
  - **Endocrine regulation** (内分泌调节): hormones

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18.2 From Mouth to Stomach

- **Mastication** (咀嚼): chewing mixes food with saliva
- **Saliva** (唾液) contains:
  - Mucus for lubrication
  - Salivary amylase for starch digestion
  - Antimicrobial agents
- **Deglutition** (吞咽): three phases—oral, pharyngeal, esophageal

18.2 From Mouth to Stomach

- **Peristalsis** (蠕动): wavelike muscle contractions propel food through the GI tract
- Circular muscle contracts behind and relaxes ahead of the bolus
- Longitudinal muscle shortens to move the bolus forward
- Progresses at 2-4 cm per second in esophagus

18.2 From Mouth to Stomach

- The **esophagus** (食管) connects pharynx to stomach
- Approximately 25 cm long
- Lower esophageal sphincter prevents regurgitation
- Food moves by peristalsis into the stomach

18.2 From Mouth to Stomach

- The **stomach** (胃) has multiple regions with specific functions:
  - **Cardia** (贲门): junction with esophagus
  - **Fundus** (胃底): upper portion for storage
  - **Body** (胃体): main digestive region
  - **Pyloric region** (幽门区): includes antrum and sphincter

18.2 From Mouth to Stomach

- Stomach functions:
  - Store food temporarily
  - Initiate protein digestion with pepsin
  - Kill bacteria with strong acidity (pH 1.5-2.5)
  - Produce chyme for small intestine

18.2 From Mouth to Stomach

- **Gastric pits** (胃小凹) are openings to gastric glands
- Multiple cell types with specialized secretions:
  - **Parietal cells** (壁细胞): secrete HCl and intrinsic factor
  - **Chief cells** (主细胞): secrete pepsinogen
  - **Mucous cells**: secrete protective mucus
  - **G cells**: secrete gastrin hormone
  - **ECL cells**: secrete histamine

18.2 From Mouth to Stomach

- **Parietal cells** (壁细胞) secrete hydrochloric acid (HCl) through an active transport mechanism
- Apical membrane: H^+-K^+ ATPase (proton pump) exchanges H^+ for K^+
- Basolateral membrane: Cl^--HCO_3^- exchanger
- $CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$

18.2 From Mouth to Stomach

- Gastric acid functions:
  - Activates pepsinogen to pepsin
  - Provides optimal pH for pepsin activity
  - Kills ingested bacteria
  - Denatures proteins for easier digestion

18.2 From Mouth to Stomach

- **Pepsinogen** (胃蛋白酶原) is the inactive enzyme precursor
- Activated to **pepsin** (胃蛋白酶) by:
  - HCl cleaving a portion of the molecule
  - Autocatalytic action: pepsin activates more pepsinogen
- Pepsin digests proteins into smaller polypeptides

18.2 From Mouth to Stomach

- Protective mechanisms prevent self-digestion:
  - Thick mucus layer forms physical barrier
  - Bicarbonate secretion neutralizes acid at epithelium
  - Tight junctions between cells prevent acid penetration
  - Rapid epithelial cell turnover (every 3 days)

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18.7 Digestion and Absorption

18.3 Small Intestine

- The **small intestine** (小肠) is the primary site for digestion and absorption
- Approximately 3 meters long, divided into three regions:
  - **Duodenum** (十二指肠): receives chyme and secretions
  - **Jejunum** (空肠): middle section for absorption
  - **Ileum** (回肠): terminal section, absorbs bile salts and vitamin B_{12}

18.3 Small Intestine

- Surface area adaptations maximize absorption:
  - Circular folds (plicae circulares)
  - Villi (fingerlike projections)
  - Microvilli (brush border)
  - Total surface area: 300 square meters

18.3 Small Intestine

- **Intestinal villi** (肠绒毛) contain:
  - Blood capillaries for absorbing amino acids and monosaccharides
  - **Lacteal** (乳糜管): lymphatic vessel for absorbing fats
  - Smooth muscle for villus contraction
- **Intestinal crypts** (肠腺): produce new epithelial cells by mitosis

18.3 Small Intestine

- **Microvilli** (微绒毛) are cytoplasmic extensions on epithelial cell surface
- Form the **brush border** (刷状缘)
- Increase surface area 20-fold
- Contain digestive enzymes embedded in their plasma membrane

18.3 Small Intestine

- **Brush border enzymes** (刷状缘酶) complete digestion
- Embedded in microvilli membrane with active sites facing lumen
- Include disaccharidases (maltase, sucrase, lactase)
- Include peptidases (aminopeptidase, enterokinase)

18.3 Small Intestine

- Brush border enzyme categories:
  - Disaccharidases break down disaccharides to monosaccharides
  - Peptidases break down small peptides to amino acids
  - Enterokinase activates pancreatic trypsinogen

18.3 Small Intestine

- **Segmentation** (分节运动): localized contractions that mix chyme
- Simultaneous contractions of multiple intestinal segments
- Mixes chyme with digestive enzymes and mucus
- Facilitates contact with absorptive surface

18.3 Small Intestine

- **Peristalsis** (蠕动): propulsive contractions move chyme along
- Intestinal motility regulated by slow waves and neural input

18.3 Small Intestine

- **Slow waves** (慢波) are spontaneous electrical rhythms
- Produced by **interstitial cells of Cajal (ICC)** (Cajal 间质细胞)
- Pace smooth muscle contractions
- When threshold reached, action potentials trigger contraction
- Much slower than cardiac pacemaker (seconds vs. milliseconds)

18.3 Small Intestine

- Electrical control of intestinal motility:
- **ICC cells** generate slow waves that pace contractions
- Smooth muscle cells respond to depolarization
- **Autonomic neurons** modulate activity through neurotransmitters
- Coordinated system ensures proper digestive function

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18.4 Large Intestine

- The **large intestine** (大肠) or **colon** (结肠) is approximately 1.5 meters long
- Primary functions:
  - Absorb water and electrolytes
  - Store fecal material
  - Host intestinal microbiota
  - Form and eliminate feces

18.4 Large Intestine

- Anatomical regions of the large intestine:
  - **Cecum** (盲肠): pouch at junction with ileum
  - **Ascending colon** (升结肠), **transverse colon** (横结肠)
  - **Descending colon** (降结肠), **sigmoid colon** (乙状结肠)
  - **Rectum** (直肠): terminal portion

18.4 Large Intestine

- **Haustra** (结肠袋): sacculations characteristic of colon
- **Taeniae coli** (结肠带): three longitudinal muscle bands

18.4 Large Intestine

- The **appendix** (阑尾) is attached to the cecum
- Contains abundant lymphatic tissue (lymphoid nodules)
- Functions in immunity and harbors beneficial bacteria
- **Appendicitis** (阑尾炎): inflammation requiring surgical removal

18.4 Large Intestine

- Radiographic visualization of large intestine after barium enema
- Clearly shows **haustra** (结肠袋): the characteristic sacculations
- Allows detection of structural abnormalities, obstructions, or tumors

18.4 Large Intestine

- **Intestinal microbiota** (肠道微生物群): trillions of beneficial bacteria
- Benefits include:
  - Synthesize vitamins K and B
  - Ferment dietary fiber
  - Produce short-chain fatty acids
  - Compete with pathogens
  - Stimulate immune system development

18.4 Large Intestine

- Fluid balance in the intestine:
  - 9 liters enter GI tract daily (2L ingested, 7L secreted)
  - Small intestine absorbs 8.5 liters
  - Large intestine absorbs most remaining 500 ml
  - Only 100 ml lost in feces

18.4 Large Intestine

- **Defecation** (排便): elimination of feces
- Controlled by two sphincters:
  - **Internal anal sphincter** (内肛门括约肌): smooth muscle, involuntary
  - **External anal sphincter** (外肛门括约肌): skeletal muscle, voluntary
- Defecation reflex triggered by rectal distension

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18.5 Liver, Gallbladder, and Pancreas

- Three major accessory organs aid digestion:
  - **Liver** (肝脏): produces bile, performs metabolic functions
  - **Gallbladder** (胆囊): stores and concentrates bile
  - **Pancreas** (胰腺): produces digestive enzymes and bicarbonate

18.5 Liver, Gallbladder, and Pancreas

- **Hepatic lobules** (肝小叶): structural units of liver
- Blood enters from periphery via portal triad:
  - **Hepatic portal vein** (肝门静脉): brings nutrient-rich blood from intestine
  - **Hepatic artery** (肝动脉): brings oxygenated blood
  - **Bile duct** (胆管): carries bile away
- Blood flows through sinusoids to central vein

18.5 Liver, Gallbladder, and Pancreas

- Blood and bile flow in opposite directions within liver lobule:
- Blood: portal triad → sinusoids → central vein (periphery to center)
- Bile: hepatocytes → bile canaliculi → bile ductules (center to periphery)
- This arrangement allows hepatocytes to process blood and secrete bile

18.5 Liver, Gallbladder, and Pancreas

- The liver performs over 500 functions, including:
  - Carbohydrate metabolism (gluconeogenesis, glycogen storage)
  - Lipid metabolism (cholesterol synthesis, lipoprotein production)
  - Protein metabolism (plasma protein synthesis, urea production)
  - Detoxification of drugs and toxins
  - Bile production
  - Storage of vitamins and minerals

18.5 Liver, Gallbladder, and Pancreas

- Major liver functions organized by category:
  - Metabolic functions: glucose homeostasis, lipid and protein metabolism
  - Synthesis: plasma proteins, clotting factors, bile
  - Detoxification: Phase I and II reactions
  - Storage: glycogen, vitamins A, D, E, K, B_{12}, iron
  - Excretion: bilirubin, drugs, hormones

18.5 Liver, Gallbladder, and Pancreas

- The liver excretes various compounds into bile:
  - Bile salts (for fat digestion, recycled via enterohepatic circulation)
  - Bile pigments (bilirubin from hemoglobin breakdown)
  - Cholesterol and lecithin
  - Drugs and toxins (after detoxification)

18.5 Liver, Gallbladder, and Pancreas

- **Enterohepatic circulation** (肠肝循环) recycles substances between liver and intestine
- Most important for bile salts: 95% reabsorbed in ileum
- Bile salts return to liver via hepatic portal vein
- Efficient recycling maintains adequate bile salt pool
- Also applies to some drugs and urobilinogen

18.5 Liver, Gallbladder, and Pancreas

- **Bilirubin metabolism** (胆红素代谢):
  - Old red blood cells broken down
  - Hemoglobin → heme + globin
  - Heme → biliverdin + CO + Fe^{2+} (iron recycled)
  - Biliverdin → bilirubin
  - Liver conjugates bilirubin with glucuronic acid
  - Conjugated bilirubin excreted into bile

18.5 Liver, Gallbladder, and Pancreas

- In the intestine, bacteria convert bilirubin to **urobilinogen** (尿胆原)
- Fates of urobilinogen:
  - Oxidized to stercobilin → brown color of feces
  - Reabsorbed and recycled to liver (enterohepatic circulation)
  - Small amount filtered by kidneys → urobilin in urine (yellow color)

18.5 Liver, Gallbladder, and Pancreas

- **Jaundice** (黄疸): yellow discoloration from elevated bilirubin
- Three types:
  - **Prehepatic** (肝前性): excessive hemolysis
  - **Hepatic** (肝性): liver disease
  - **Posthepatic** (肝后性): bile duct obstruction

18.5 Liver, Gallbladder, and Pancreas

- **Bile acids** (胆汁酸) are amphipathic molecules
- Have both polar (hydrophilic) and nonpolar (hydrophobic) regions
- In water, aggregate to form **micelles** (微团)
- Micelles incorporate cholesterol and lecithin
- Function: emulsify triglycerides in chyme for digestion

18.5 Liver, Gallbladder, and Pancreas

- The **pancreas** (胰腺) has both endocrine and exocrine functions
- Pancreatic secretions flow through duct system
- Pancreatic duct joins common bile duct
- Both empty into duodenum at ampulla of Vater
- Controlled by sphincter of Oddi

18.5 Liver, Gallbladder, and Pancreas

- **Gallstones** (胆结石) or biliary calculi
- Solid deposits that form in gallbladder
- Types: cholesterol stones (most common), pigment stones
- Can cause pain, obstruction, infection
- Treatment may include cholecystectomy (surgical removal)

18.5 Liver, Gallbladder, and Pancreas

- The pancreas has distinct endocrine and exocrine portions:
- **Endocrine** (内分泌): pancreatic islets secrete insulin and glucagon
- **Exocrine** (外分泌): acinar cells produce digestive enzymes
- Acinar cells store inactive enzymes in zymogen granules
- Enzymes secreted into duct system → duodenum

18.5 Liver, Gallbladder, and Pancreas

- Pancreatic duct cells secrete **bicarbonate** (碳酸氢盐) into pancreatic juice
- Mechanism:
  - CO_2 from blood + $H_2O \rightarrow H_2CO_3$
  - $H_2CO_3 \rightarrow HCO_3^-$ + H^+
  - HCO_3^- secreted via Cl^- exchanger
  - Cl^- leaks back through CFTR chloride channels
- Function: neutralizes acidic chyme in duodenum

18.5 Liver, Gallbladder, and Pancreas

- Pancreatic juice contains powerful digestive enzymes:
  - Proteases: trypsin, chymotrypsin, carboxypeptidase, elastase
  - **Pancreatic amylase** (胰淀粉酶): digests starch
  - **Pancreatic lipase** (胰脂肪酶): digests triglycerides
  - Nucleases: digest RNA and DNA

18.5 Liver, Gallbladder, and Pancreas

- Proteolytic enzymes secreted as inactive **zymogens** (酶原) to prevent autodigestion
- **Trypsinogen** (胰蛋白酶原) activated by brush border enzyme **enterokinase** (肠激酶)
- Active **trypsin** (胰蛋白酶) then activates other zymogens
- Cascade mechanism ensures enzymes activate only in intestinal lumen

18.5 Liver, Gallbladder, and Pancreas

- **Cystic fibrosis** (囊性纤维化): genetic defect in CFTR chloride channel
- Affects pancreatic juice bicarbonate secretion
- Causes thick mucus accumulation
- Results in pancreatic insufficiency and malabsorption

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18.7 Digestion and Absorption

18.6 Neural and Endocrine Regulation

- Digestive regulation coordinates secretion and motility
- Mechanisms include:
  - Neural control: autonomic and enteric nervous systems
  - Hormonal control: gastrin, CCK, secretin, GIP
  - Paracrine control: histamine, somatostatin

18.6 Neural and Endocrine Regulation

- Gastric secretion occurs in three phases:
  - **Cephalic phase** (脑相): before food enters stomach (30%)
  - **Gastric phase** (胃相): when food is in stomach (60%)
  - **Intestinal phase** (肠相): when chyme enters duodenum (10%)

18.6 Neural and Endocrine Regulation

- Regulation of **gastric acid secretion** (胃酸分泌):
- Pathway involves three cell types:
  - **G cells** secrete gastrin (stimulated by vagus nerve and peptides)
  - **ECL cells** release histamine (stimulated by gastrin)
  - **Parietal cells** secrete HCl (stimulated by histamine, gastrin, ACh)
- Negative feedback: low pH stimulates D cells to secrete somatostatin

18.6 Neural and Endocrine Regulation

- **Gastrin** (胃泌素): stimulates gastric acid and pepsin secretion
- **Histamine** (组胺): paracrine regulator, strongly stimulates parietal cells
- **Acetylcholine** (乙酰胆碱): from vagus nerve, stimulates all gastric secretions
- **Somatostatin** (生长抑素): inhibits gastrin and histamine release

18.6 Neural and Endocrine Regulation

- Gastrointestinal hormones and their effects:
  - **Gastrin**: stimulates gastric secretion and motility
  - **CCK** (胆囊收缩素): stimulates pancreatic enzymes, gallbladder contraction
  - **Secretin** (促胰液素): stimulates pancreatic bicarbonate, bile secretion
  - **GIP** (胃抑制肽): inhibits gastric secretion, stimulates insulin

18.6 Neural and Endocrine Regulation

- **Cholecystokinin (CCK)**: released in response to fat and protein
- Effects include:
  - Stimulates pancreatic enzyme secretion
  - Causes gallbladder contraction
  - Relaxes sphincter of Oddi
  - Inhibits gastric emptying
  - Promotes satiety

18.6 Neural and Endocrine Regulation

- **Secretin**: released in response to acid in duodenum
- Effects include:
  - Stimulates pancreatic bicarbonate secretion
  - Stimulates bile secretion
  - Inhibits gastric acid secretion and motility
  - Protects duodenum from excessive acid

18.6 Neural and Endocrine Regulation

- The **enteric nervous system** (肠神经系统) functions as an "enteric brain"
- Contains motor neurons, interneurons, and sensory neurons
- Coordinates peristalsis through local reflexes
- Can function independently but modulated by autonomic nervous system

18.6 Neural and Endocrine Regulation

- Peristaltic reflex (“law of the intestine”):
- Distension of intestine triggers coordinated response:
  - Oral side (above bolus): contraction (ACh, substance P)
  - Anal side (below bolus): relaxation (NO, VIP)
- Sensory neurons detect stimulus
- Interneurons process information
- Motor neurons coordinate muscle response

18.6 Neural and Endocrine Regulation

- **Trophic effects** (营养效应): hormones promote tissue growth
- **Gastrin**: stimulates growth of gastric mucosa
- **CCK**: stimulates growth of pancreatic acini
- **Secretin**: stimulates growth of pancreatic ducts
- Important for maintaining digestive system structure

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18.7 Digestion and Absorption

18.7 Digestion and Absorption

- Digestion breaks down macromolecules into absorbable units
- Absorption transfers nutrients into blood or lymph
- Different mechanisms for:
  - Carbohydrates (monosaccharides)
  - Proteins (amino acids)
  - Lipids (fatty acids, monoglycerides)

18.7 Digestion and Absorption

- Major digestive enzymes and their characteristics:
  - Site of secretion (saliva, gastric juice, pancreatic juice, brush border)
  - Substrate specificity
  - Products generated
  - Optimal pH for activity

18.7 Digestion and Absorption

- **Starch digestion** (淀粉消化):
- Begins in mouth with salivary amylase (minimal)
- Main digestion by pancreatic amylase in small intestine
- Produces maltose, maltriose, and α-limit dextrins
- Final digestion by brush border enzymes

18.7 Digestion and Absorption

- **Pancreatic amylase** (胰淀粉酶) action on starch:
- Cleaves α-1,4 glycosidic bonds within starch chains
- Cannot cleave α-1,6 branch points
- Products: maltose (2 glucose units), maltriose (3 glucose units), α-limit dextrins (branched oligosaccharides)
- Brush border enzymes complete digestion to glucose

18.7 Digestion and Absorption

- Carbohydrate absorption:
- **Glucose** and **galactose**: SGLT1 (Na^+-glucose cotransporter)
- **Fructose**: GLUT5 (facilitated diffusion)
- All exit into blood via GLUT2 at basolateral membrane
- Enter hepatic portal vein → liver

18.7 Digestion and Absorption

- **Protein digestion** (蛋白质消化):
- Begins in stomach with pepsin
- Main digestion by pancreatic proteases: trypsin, chymotrypsin, elastase, carboxypeptidase
- Final digestion by brush border peptidases
- Products: amino acids, dipeptides, tripeptides

18.7 Digestion and Absorption

- Protein absorption:
- **Free amino acids**: Na^+-amino acid cotransporters (secondary active transport)
- **Dipeptides and tripeptides**: H^+-peptide cotransporter (PepT1)
- Inside cells, peptides hydrolyzed to amino acids
- Exit into blood via facilitated diffusion
- Enter hepatic portal vein → liver

18.7 Digestion and Absorption

- Lipid digestion presents unique challenges:
- Lipids are hydrophobic (water-insoluble)
- Require emulsification for enzyme access
- Products must be solubilized for transport
- Solution: bile salts and micelle formation

18.7 Digestion and Absorption

- **Pancreatic lipase** (胰脂肪酶) digests triglycerides
- Cleaves fatty acids at positions 1 and 3 of glycerol backbone
- Products: 2-monoglyceride + two free fatty acids
- Requires **colipase** (辅脂酶) to anchor to lipid droplets

18.7 Digestion and Absorption

- **Fat emulsification** (脂肪乳化) increases surface area
- Bile salts and phospholipids break large lipid droplets into smaller ones
- Provides greater access for pancreatic lipase
- Essential first step in fat digestion

18.7 Digestion and Absorption

- **Fat absorption** (脂肪吸收) pathway:
- Fatty acids and monoglycerides diffuse from micelles into cells
- Inside cells: resynthesized into triglycerides
- Combined with proteins, cholesterol, phospholipids → **chylomicrons** (乳糜微粒)
- Chylomicrons enter lacteals (lymphatic vessels)
- Lymph → thoracic duct → left subclavian vein → blood

18.7 Digestion and Absorption

- Bile salts remain in intestinal lumen
- 95% reabsorbed in terminal ileum
- Returned to liver via hepatic portal vein
- Recycled via enterohepatic circulation
- Fat-soluble vitamins (A, D, E, K) absorbed with fats

18.7 Digestion and Absorption

- **Lipoproteins** (脂蛋白) transport lipids in blood:
  - **Chylomicrons** (乳糜微粒): transport dietary fats from intestine
  - **VLDL** (极低密度脂蛋白): transport triglycerides from liver
  - **LDL** (低密度脂蛋白): transport cholesterol to tissues (“bad cholesterol”)
  - **HDL** (高密度脂蛋白): transport cholesterol to liver (“good cholesterol”)

18.7 Digestion and Absorption

- Lipoprotein characteristics differ in:
  - Size and density
  - Lipid and protein composition
  - Origin and destination
  - Function in lipid transport
- LDL/HDL ratio important for cardiovascular health

18.7 Digestion and Absorption

- Clinical applications:
- **Proton pump inhibitors (PPIs)** (质子泵抑制剂): block H^+-K^+ ATPase
- **H_2 blockers** (H_2受体阻断剂): block histamine receptors
- **Antacids** (抗酸剂): neutralize stomach acid
- All reduce gastric acidity for GERD and ulcers

18.7 Digestion and Absorption

- Key concepts for exam preparation:
- Understand the layered structure of GI tract
- Know cell types and their secretions
- Trace the path of nutrients from ingestion to absorption
- Explain hormonal and neural regulation
- Describe enzyme activation cascades
- Understand lipid digestion and transport